Worksheet for Lecture 8

Problem 1: IoU Calculation

Given the following bounding boxes:
- Ground truth box: $(x, y, w, h) = (1, 1, 3, 6)$.
- Predicted box: $(\hat{x}, \hat{y}, \hat{w}, \hat{h}) = (2, 2, 5, 5)$.

Both (x, y) and $(\hat{x}, \hat{y})$ represent the coordinates of the bottom-left corner of the bounding box.

Calculate the Intersection over Union (IoU) between the two bounding boxes.

Problem 2: Bounding Box Loss Calculation

Given the following:
- Predicted bounding box: $(x_1, y_1, x_2, y_2) = (2, 2, 5, 5)$.
- Predicted class probabilities: $[0.8, 0.2]$ for class 1 and class 2.
- Ground truth bounding box: $(x_1, y_1, x_2, y_2) = (1, 1, 4, 4)$.
- Ground truth class label: class 1.

a) Calculate the loss between the predicted bounding box and ground truth bounding box using the L_1 and L_{iou} loss.
b) Calculate the classification loss using cross-entropy loss for the predicted class probabilities, assume the $\lambda_{iou} = 1$ and $\lambda_{L1} = 1$.

Problem 3: Hungarian Matching

Given the following cost matrix for 4 predicted bounding boxes (columns) and 4 ground truth bounding boxes (rows):

$$\begin{matrix} 0.1 & 0.5 & 0.2 & 0.7 \\ 0.2 & 0.6 & 0.3 & 0.1 \\ 0.4 & 0.3 & 0.5 & 0.6 \\ 0.8 & 0.2 & 0.4 & 0.3 \end{matrix}$$

where the rows correspond to the ground truth boxes and the columns correspond to the predicted boxes.

Apply the Hungarian algorithm to match the predictions to the ground truth boxes, give the intermediate steps and compute the optimal total cost.

Problem 4: Max Pooling Operation

Given the following 5x5 matrix:

$$\begin{matrix} 9 & 21 & 23 & 3 & 20 \\ 1 & 22 & 14 & 4 & 8 \\ 17 & 19 & 5 & 12 & 10 \\ 15 & 7 & 6 & 25 & 24 \\ 16 & 11 & 2 & 13 & 18 \end{matrix}$$

Perform max pooling with a 3x3 kernel, stride=2, and padding=1. Show the output.

Problem 5: Deconvolution Operation

Given the following 2x2 matrix:

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix}$$

and a 3x3 convolution kernel:

$$\begin{matrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{matrix}$$

Perform a deconvolution operation (transpose convolution) with stride=2 and padding=1. Show the padded matrix and the final output.